

Quiz 6

Tick the statements that are true and leave unticked those that are false.

Each statement is worth 1 point for a correct answer.

Your name: _____

Reminder: We represent a pair of A and B as $\lambda f.f AB$ and denote it (A, B) . The pair constructor is $\text{PAIR} = \lambda x.\lambda y.\lambda f.f xy$.

- ☒ The lambda term PAIR is in normal form.
- ☐ The lambda term $\text{PAIR } AB$ is in normal form.
- ☐ $\text{PAIR } \overline{2} \overline{3} =_{\beta} \text{PAIR } \overline{3^2}$.
- ☐ $\text{PAIR}(\text{PAIR } AB)C =_{\beta} \text{PAIR } A(\text{PAIR } BC)$.
- ☒ PAIR PAIR is a valid lambda term.
- ☒ PAIR PAIR PAIR is a valid lambda term.
- ☒ $\text{PAIR } \overline{3} \overline{4} \text{ ADD} \rightarrow_{\beta}^* \overline{7}$.
- ☒ Let $\text{ISEMPTY} = \lambda l.l \text{ TRUE } (\lambda h.\lambda t.\text{FALSE})$. Then $\text{ISEMPTY } L$ correctly tests whether the Scott list L is empty.
- ☐ The term $F = \lambda x.\lambda y.\text{ISZERO } x y (F(\text{PRED } x)y)$ is a valid lambda term.
- ☒ Let Θ be the term supplied by the fixpoint theorem. Then $\Theta \text{ PAIR } x \rightarrow_{\beta}^* (\Theta \text{ PAIR}, x)$.

Bonus 1. Define a term CROSS that takes a pair (f, g) of functions and a pair (x, y) of arguments, and returns the pair (fy, gx) .

Answer: $\text{CROSS} = \lambda p.\lambda q.\text{PAIR } ((\text{FST } p)(\text{SND } q)) ((\text{SND } p)(\text{FST } q))$.

Bonus 2. Using Θ , ISZERO , PRED , and MULT , define a term E such that $E \overline{n} \rightarrow_{\beta}^* \overline{2^n}$ for every $n \geq 0$. Note that $2^0 = 1$.

Answer: $E = \Theta G$, where $G = \lambda r.\lambda n.\text{ISZERO } n \overline{1} (\text{MULT } \overline{2} (r(\text{PRED } n)))$.

Would you rather program a semester-long project on a Turing machine or in lambda calculus?

Explanations

- ⊗ *The lambda term PAIR is in normal form.* This is the term $\lambda x.\lambda y.\lambda f.fxy$. It contains no beta-redex.
- *The lambda term PAIR AB is in normal form.* Application is left-associative: $\text{PAIR } A B = (\text{PAIR } A)B$. Its first application is a beta-redex, regardless of A and B .
- $\text{PAIR } \bar{2} \bar{3} =_{\beta} \text{PAIR } \bar{3}^2$. The left side is a completed pair: $\text{PAIR } \bar{2} \bar{3} \rightarrow_{\beta}^* \lambda f.f \bar{2} \bar{3}$. The right side has only one component: $\text{PAIR } \bar{9} \rightarrow_{\beta}^* \lambda y.\lambda f.f \bar{9} y$. These are different normal forms.
- $\text{PAIR}(\text{PAIR } AB)C =_{\beta} \text{PAIR } A(\text{PAIR } BC)$. On the left, the first component is $\text{PAIR } A B$; on the right, it is A .
- ⊗ *PAIR PAIR is a valid lambda term.* A lambda term may be supplied as an argument to another lambda term. Here PAIR is both the function and its argument.
- ⊗ *PAIR PAIR PAIR is a valid lambda term.* Application is left-associative, so this means $(\text{PAIR PAIR})\text{PAIR}$. Both applications are allowed.
- ⊗ $\text{PAIR } \bar{3} \bar{4} \text{ ADD} \rightarrow_{\beta}^* \bar{7}$. The pair passes its components to ADD: $\text{PAIR } \bar{3} \bar{4} \text{ ADD} \rightarrow_{\beta}^* \text{ADD } \bar{3} \bar{4} \rightarrow_{\beta}^* \bar{7}$.
- ⊗ *Let $\text{ISEMPTY} = \lambda l.l \text{ TRUE } (\lambda h.\lambda t.\text{FALSE})$. Then $\text{ISEMPTY } L$ correctly tests whether the Scott list L is empty.* A Scott list chooses between an empty-case value and a nonempty-case function: $\text{ISEMPTY NIL} \rightarrow_{\beta}^* \text{TRUE}$, while $\text{ISEMPTY}(\text{CONS } h t) \rightarrow_{\beta}^* \text{FALSE}$.
- *The term $F = \lambda x.\lambda y.\text{ISZERO } x y (F(\text{PRED } x)y)$ is a valid lambda term.* The right-hand contains F , which we are trying to define as the right-hand side itself.
- ⊗ *Let Θ be the term supplied by the fixpoint theorem. Then $\Theta \text{PAIR } x \rightarrow_{\beta}^* (\Theta \text{PAIR}, x)$.* The fixpoint theorem gives $\Theta \text{PAIR} \rightarrow_{\beta}^* \text{PAIR } (\Theta \text{PAIR})$. Applying the latter to x gives $\text{PAIR } (\Theta \text{PAIR}) x \rightarrow_{\beta}^* (\Theta \text{PAIR}, x)$.

Bonus 1. Let $p = \text{PAIR } f g$ and $q = \text{PAIR } x y$. The projections give $\text{FST } p \rightarrow_{\beta}^* f$, $\text{SND } p \rightarrow_{\beta}^* g$, $\text{FST } q \rightarrow_{\beta}^* x$, and $\text{SND } q \rightarrow_{\beta}^* y$. Hence $\text{CROSS } p q$ reduces to $\text{PAIR } (f y) (g x)$.

Bonus 2. The body G receives the recursive function as r . It returns $\bar{1}$ at zero; otherwise it doubles the result on the predecessor. With $E = \Theta G$, unfolding the fixpoint gives $E \rightarrow_{\beta}^* GE$. Thus $E \bar{0} \rightarrow_{\beta}^* \bar{1}$ and $E n + 1 \rightarrow_{\beta}^* \text{MULT } \bar{2} (E \bar{n}) \rightarrow_{\beta}^* \overline{2^{n+1}}$.